Science — Unit 4, electricity and thunderstorms · Grade 6

Storm Lab: The Weather Station

Answer the 12 questions. Use the observations given; you do not need to perform an experiment. Allow 10 to 15 minutes.

> You are reviewing a weather station's training cards. Practise conductors, insulators, experiments, and thunderstorm safety.

1 What is the name of the tall storm clouds discussed in Unit 4? Write the cloud name only.

2 For flash A, you hear thunder 3 seconds later. For flash B, you hear thunder 9 seconds later. Which flash was farther away? Write A or B.

3 Why do you usually see a flash before hearing its thunder?

- Light travels faster than sound.
- Sound travels faster than light.
- The flash waits for the thunder.

4 What helps separate electrical charges inside a storm cloud?

- The cloud touching a metal roof
- Collisions between small ice crystals and soft hail
- Two whole clouds rubbing together like balloons

5 Why does a typical electrical cable have a metal wire inside a plastic cover?

- Both parts stop current equally.
- Plastic carries current and metal blocks it.
- Metal carries current; plastic helps insulate it.

6 Two similar pieces of wood are tested. One is dry; the other is wet with ordinary tap water. Which would generally conduct better?

- The wet piece
- The dry piece
- Moisture can never affect wood

7 Which sample would generally conduct electricity better: very pure water or water with dissolved table salt?

- Very pure water
- Water with dissolved salt
- Neither can carry any current

8 In a working battery-and-bulb test, inserting a metal key makes the bulb light. What does this result show?

- All materials conduct equally well.
- The key creates electricity by itself.
- The key allows enough current to flow to light the bulb.

9 Air normally insulates well. How can an electrical discharge still pass through it in a storm?

- A very strong electric field can make the air conduct.
- Air is always as good a conductor as copper.
- Air conducts only when every metal object is removed.

10 You hear thunder outside. Which is the appropriate shelter among these choices?

- A picnic shelter with open sides
- Inside a solid building with closed walls and a roof
- Under a lone tall tree

11 The storm seems to have passed. How long should you wait after the last thunder before returning outdoors?

- Until the rain stops
- Just 2 minutes
- At least 30 minutes

12 What is the purpose of a properly installed lightning protection system on a building?

- Provide a conducting route to the ground if the building is struck
- Pull every nearby storm toward the roof
- Store the storm inside a metal rod

Generated from the sources listed in the quiz definition file.